Registers and Counters - Assignment

1 Design a counter with T flip flops whose count sequence is 0, 2, 4, 5, 3, 1, 0.....

Let

Current state			next state					
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	1	0	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
1	1	0	x	x	x	x	x	x
1	1	1	x	x	x	x	x	x

$T_2 =$

$Q_2 \backslash Q_1Q_0$	00	01	11	10
0	0	1	0	0
1	1	0	X	X

$$T_2 = Q_2\overline{Q_0} + \overline{Q_2}\,\overline{Q_1}\,Q_0$$

$T_1 =$

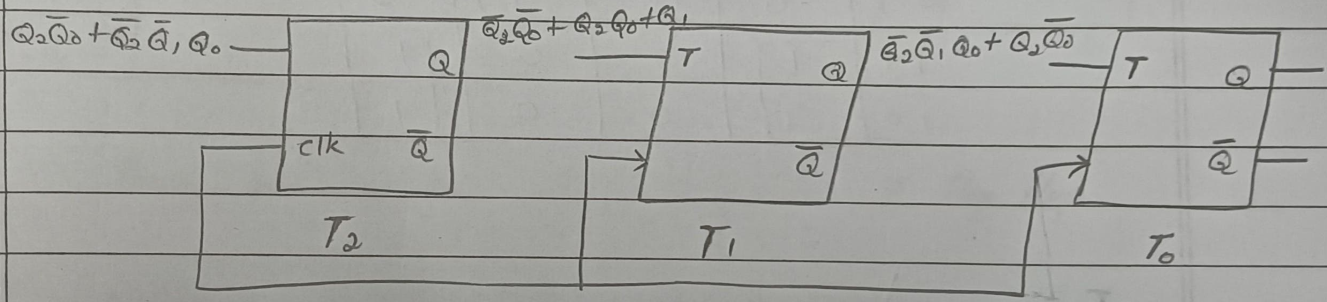
$Q_2 \backslash Q_1Q_0$	00	01	11	10
0	1	0	1	1
1	0	1	X	X

$$T_1 = \overline{Q_2}\,\overline{Q_0} + Q_2Q_0 + Q_1$$

$T_0 =$

$Q_2 \backslash Q_1Q_0$	00	01	11	10
0	0	1	0	0
1	1	0	X	X

$$T_0 = \overline{Q_2}\,\overline{Q_1}\,Q_0 + Q_2\overline{Q_0}$$



2 Design a 3 bit up/down counter using negative edge triggered T-FF.

Input	Present state			next state			T flip-flop-input		
M	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1

0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

$T_0 = 1$

MQ_2 \ Q_1Q_0	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$\therefore T_1 = \bar{M}Q_0 + M\bar{Q_0}$

$T_2 =$

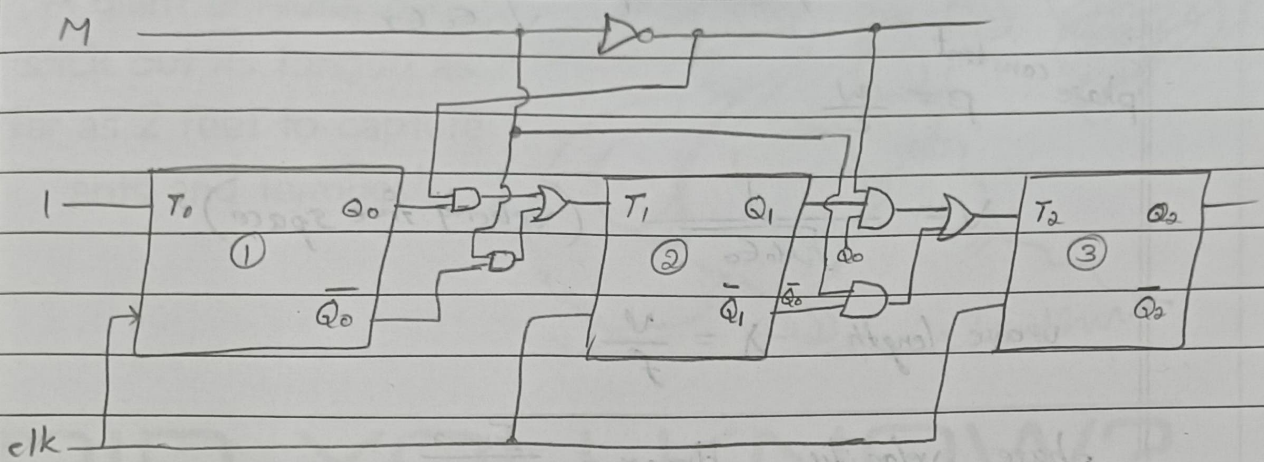
MQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

$T_2 = \bar{M}Q_1Q_0 + M\bar{Q_1}\bar{Q_0}$

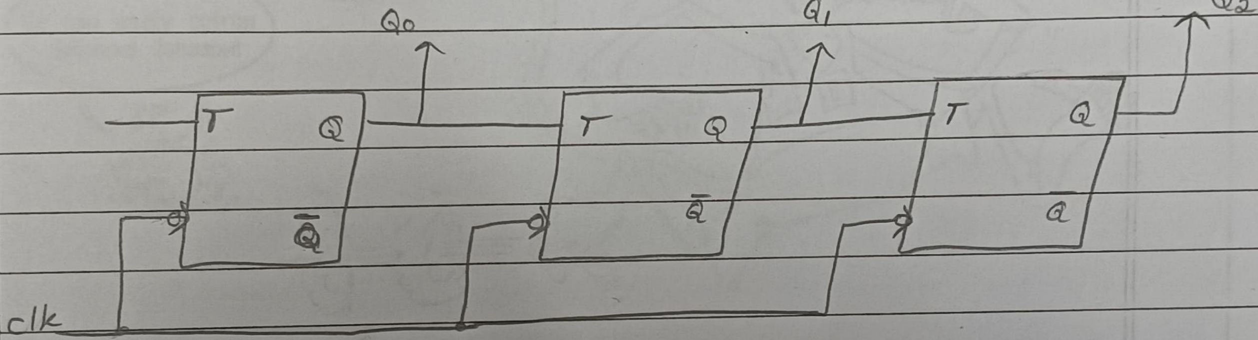
$\therefore T_0 = 1$

$T_1 = \bar{M}\,Q_0 + M\bar{Q_0}$

$T_2 = \bar{M}\,Q_1Q_0 + M\bar{Q_1}\,\bar{Q_0}$



3 The circuit in the figure below is a counter. What is the sequence that this circuit counts in?



These are negative edge triggered T flip flop

assuming initial state $Q_0 Q_1 Q_2 = 000$

sequence	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	1	1	1
4	0	0	0
5	0	0	1
6	0	1	0

∴ The sequence it follows is $000 \to 001 \to 010 \to 111 \to 000 \ldots$